Fundamentals of Computer Design

(Computer Architecture: Chapter 1)

Yanyan Shen

Department of Computer Science and Engineering

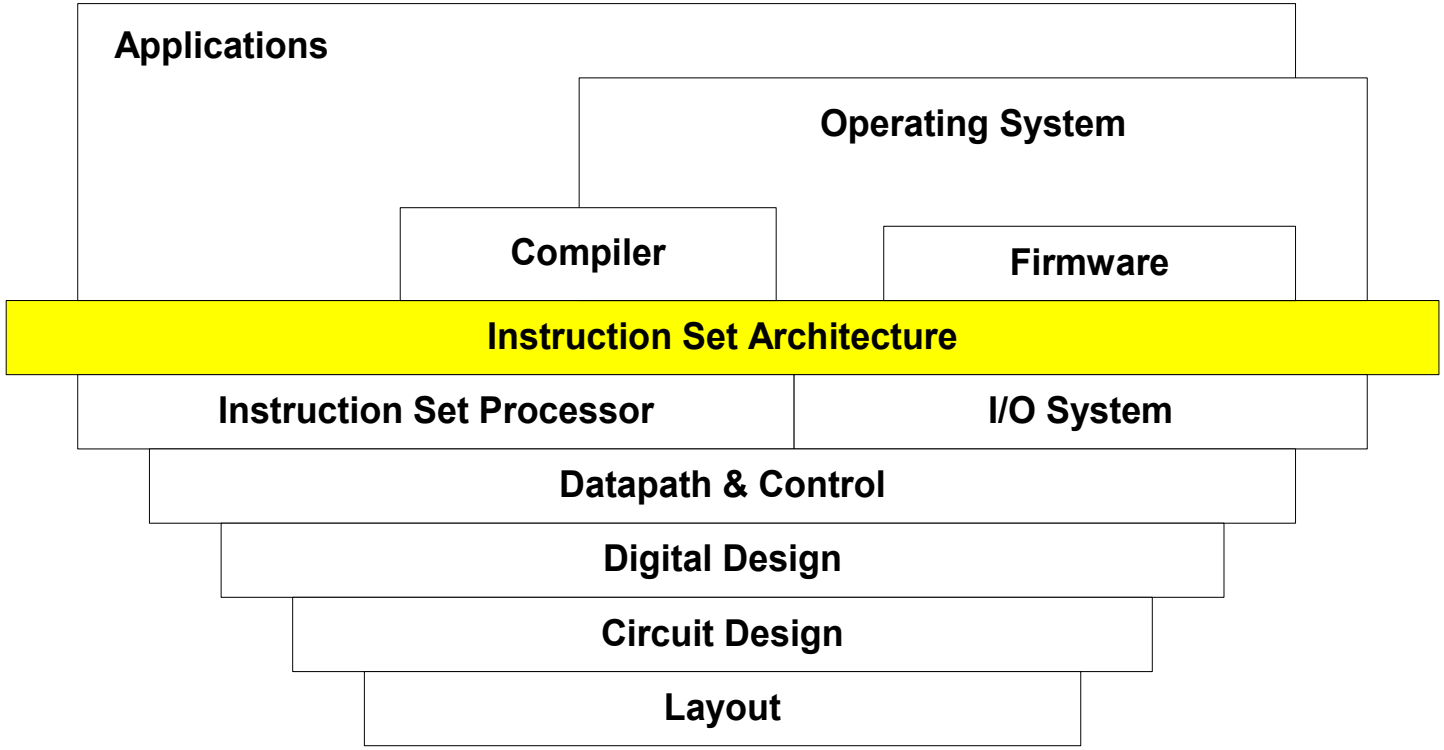
Agenda

- Introduction
- Classes of Computers
- 1.3 Defining Computer Architecture
- Trends in Technology
- Trends in Power and Energy in ICs
- Trends in Cost
- Dependability
- Measuring Performance
- Quantitative Principles

Defining Computer Architecture

- "Old" view of computer architecture:
  - Instruction Set Architecture (ISA) design
  - i.e., decisions regarding:
    - registers, memory addressing, addressing modes, instruction operands, available operations, control flow instructions, instruction encoding
- The ISA is roughly the same as the programming model of a processor as seen by an assembly language programmer or compiler writer.

Instruction Set Architecture



S/W and H/W consists of hierarchical layers of abstraction, each hides details of lower layers from the above layer

The instruction set arch. abstracts the H/W and S/W interface and allows many implementation of varying cost and performance to run the same S/W

Contents of Instruction Set Architecture

- Registers
- Class of ISA
- Memory addressing
  - Byte addressing
- Addressing modes
- Types and sizes of operands
- Operations
- Control flow instructions
- Encoding an ISA
  - Fixed length and variable length

Addressing Modes

Example Instruction	Meaning	When used
Register	Add R4,R3	R4 <- R4 + R3
Immediate	Add R4, #3	R4 <- R4 + 3
Displacement	Add R4, 100(R1)	R4 <- R4 + M[100+R1]
Register deferred	Add R4,(R1)	R4 <- R4 + M[R1]
Indexed	Add R3, (R1 + R2)	R3 <- R3 + M[R1+R2]
Direct	Add R1, (1001)	R1 <- R1 + M[1001]
Memory deferred	Add R1, @(R3)	R1 <- R1 + M[M[R3]]
Auto-increment	Add R1, (R2)+	R1 <- R1 +M[R2] R2 <- R2 + d
Auto-decrement	Add R1,-(R2)	R2 <-R2-d R1 <- R1 + M[R2]
Scaled	Add R1, 100(R2)[R3]	R1<-R1+M[100+R2+R3*d]

Example ISAs

- ARM
  - ARMv1-v7
  - ARMv7 extensions
    - Thumb-2
    - NEON – media acceleration technology[8]
    - VFP v3
- HP
  - HP 2100
  - PA-RISC
- IBM
  - IBM 8100
  - IBM Series
  - Power Architecture
    - POWER
    - PowerPC
    - PowerPC AS
- MIPS

- Intel
  - IA-64
  - X86
  - x86 extensions
    - FPU (x87) – Floating-point-unit (FPU) instructions
    - MMX – MMX SIMD instructions
    - MMX Extended – extended MMX SIMD instructions
    - SSE – streaming SIMD extensions (SSE) instructions (70 instructions)
    - SSE2 – streaming SIMD extensions 2 instructions (144 new instructions)
    - SSE3 – streaming SIMD extensions 3 instructions (13 new instructions)
    - SSSE3 – supplemental streaming SIMD extensions (16 instructions)
    - SSE4.1 – streaming SIMD extensions 4, Penryn subset (47 instructions)
    - SSE4.2 – streaming SIMD extensions 4, Nehalem subset (7 instructions)
    - SSE4 – All streaming SIMD extensions 4 instructions (both SSE4.1 and SSE4.2)
    - SSE4a – streaming SIMD extensions 4a (AMD)
    - SSE5 – streaming SIMD extensions 5 (170 instructions)
    - XSAVE – XSAVE instructions
    - AVX – advanced vector extensions instructions
    - FMA – fused multiply-add instructions
    - AES – Advanced Encryption Standard instructions
    - CLMUL – Carry-less mtiply (PCLMULQDQ) instruction
    - 3DNow![citation needed] – 3DNow! instructions (21 instructions)
    - 3DNow! Extended – extended 3DNow! instructions (5 instructions)
    - Cyrix – Cyrix-specific instructions
    - AMD – AMD-specific instructions (older than K6)
    - SMM – System management mode instructions
    - SVM – Secure virtual machine instructions
    - PadLock – VIA PadLock instructions

MIPS Instruction Fields

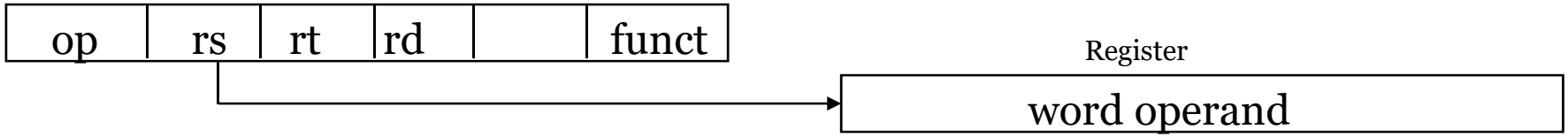
- MIPS fields are given names to make them easier to refer to

op	rs	rt	rd	shamt	funct

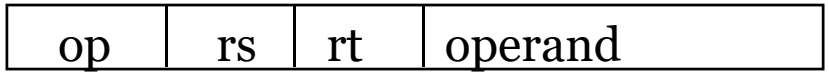
op	6-bits	opcode that specifies the operation
rs	5-bits	register file address of the first source operand
rt	5-bits	register file address of the second source operand
rd	5-bits	register file address of the result's destination
shamt	5-bits	shift amount (for shift instructions)
funct	6-bits	function code augmenting the opcode

MIPS Addressing Modes Examples

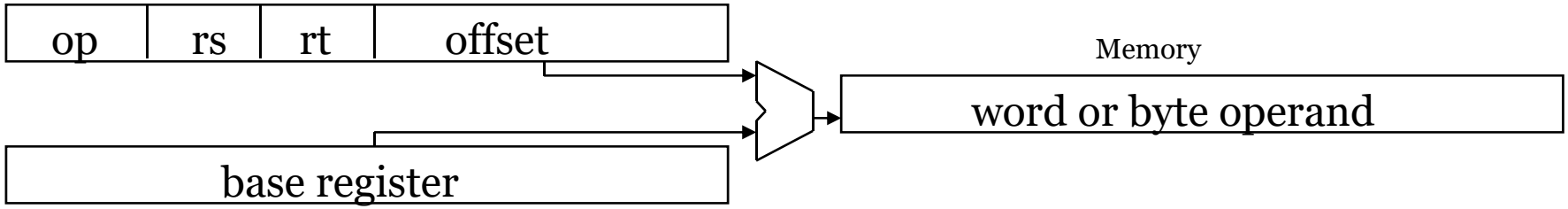
1. Register addressing



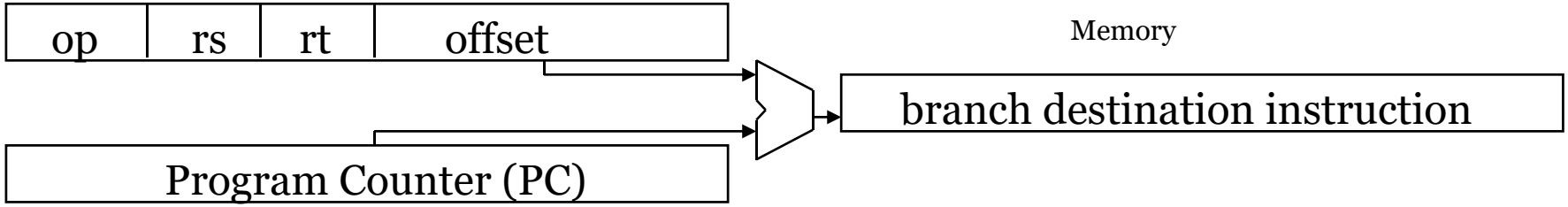
2. Immediate addressing



3. Base (displacement) addressing

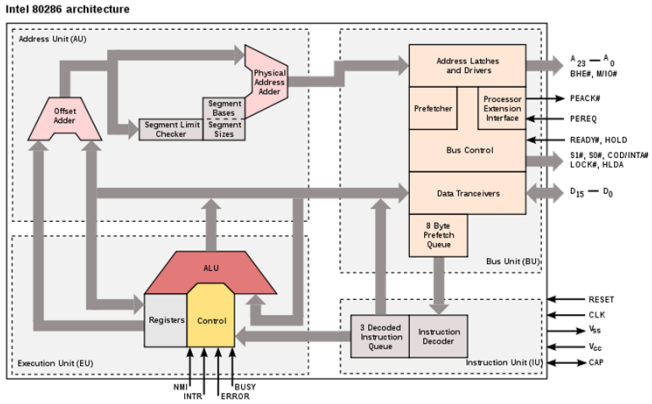


4. PC-relative addressing



Microarchitecture

- **microarchitecture**, also called **computer organization**, is the way a given ISA is implemented on a processor.
  - A given ISA may be implemented with different microarchitectures.
- The microarchitecture includes the constituent parts of the processor and how these interconnect and interoperate to implement the ISA.



Intel i80286 microarchitecture

Microarchitectural Concepts

- Instruction pipelining
- Hierarchical memory organization
- Cache
- Cache coherence
- Branch prediction
- Superscalar
- Out-of-order execution
- Register renaming
- Multiprocessing and multithreading

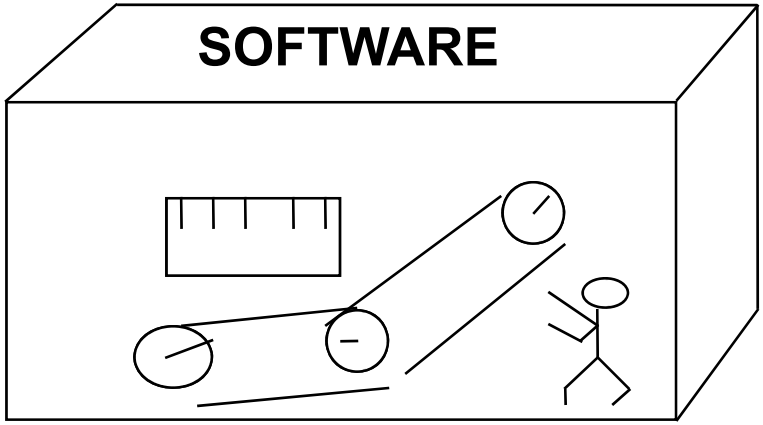
Computer Architecture in General

When we construct a building numerous practical considerations need to be taken into account:

- Available materials
- Worker skills
- Budget
- Space

Similarly, Computer Architecture is about working within constraints:

- What will the market buy?
- Cost/Performance
- Tradeoffs in materials and processes



Genuine Computer Architecture

- **Genuine definition**: *Designing the organization and hardware to meet goals and functional requirements*
- The implementation of a computer involves two components
  - 1) **organization** (or microarchitecture)
  - 2) **hardware**: detailed logic design and packaging
- In this course, **computer architecture** covers all the three:
  - (1) ***ISA***,
  - (2) ***microarchitecture*** *or* ***organization***, and
  - (3) ***hardware***

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Rapid Changes in Technology

- If an ISA is successful, it should survive rapid changes in *implementation technology*
- The designer of a computer should be aware of such rapid changes

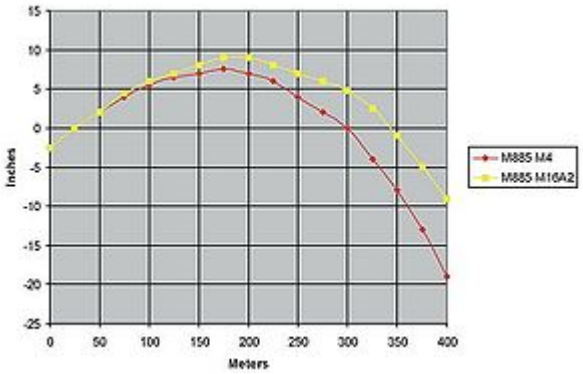
- **Five implementation technologies** are critical to model implementations of computer design

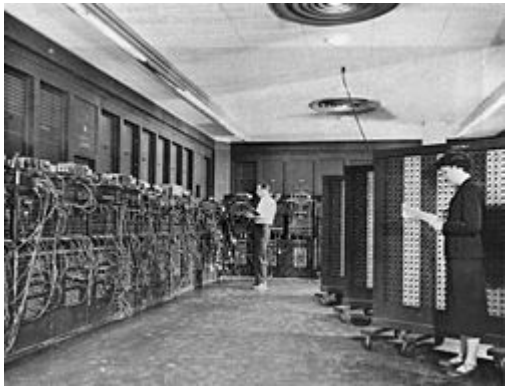
Five Critical Implementation Techs

Integrated circuit logic technology	• As Moore's law, the transistors count doubles every 18-24 months
Semiconductor DRAM	• The capacity per DRAM chip doubles every two to three years
Semiconductor Flash	• Capacity doubles roughly every two years
Magnetic disk technology	• Capacity doubles every three years
Network technology	• Bandwidth increases quickly

ENIAC - background

- Electronic Numerical Computer
- Eckert and Mauchly, University of Pennsylvania
- Purpose: *Trajectory tables for weapons*
- Started 1943, Finished 1946
  - Too late for war effort
  - Used until 1955





John Mauchly (1907-1980) &
J. Presper Eckert (1919-1995)

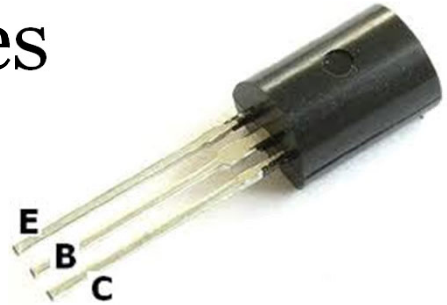
ENIAC - details

- Decimal (not binary)
- 20 accumulators of 10 digits
- Programmed manually by switches
- 18,000 **vacuum tubes**
- 30 tons
- 15,000 square feet
- 140 kW power consumption
- 5,000 additions per second

Straight ring/Overbeck counter				
State	Q0	Q1	Q2	Q3
0	1	0	0	0
1	0	1	0	0
2	0	0	1	0
3	0	0	0	1
0	1	0	0	0
1	0	1	0	0
2	0	0	1	0
3	0	0	0	1
0	1	0	0	0

Transistors – the 2nd generation

- Replaced vacuum tubes
- Advantages
  - Smaller
  - Cheaper
  - Less heat dissipation
- Solid state device
- Made from Silicon (Sand)
- Invented 1947 at **Bell Labs**
- William Shockley et al.



First working transisto

John Bardeen, William Shockley and Walter Brattain at Bell Labs, 1948.

Transistor Based Computers

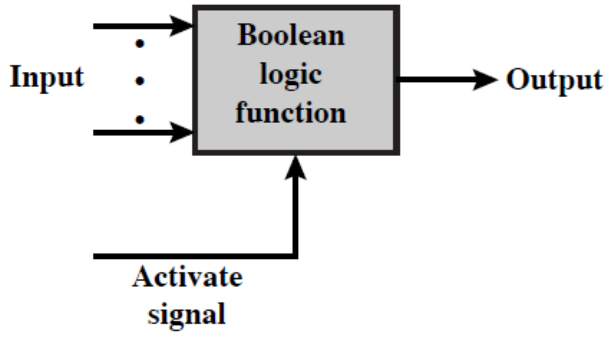
- Second generation machines
- NCR & RCA produced small transistor machines
- IBM
  - Produced IBM 7000
- DEC - 1957
  - Produced PDP-1

IBM 7000

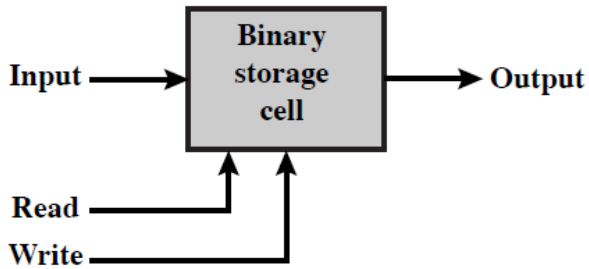
PDP-1

Microelectronics – 3rd Generation

- Literally - "small electronics"
- A computer is made up of gates, memory cells and interconnections
- These can be manufactured on a **<u>semiconductor</u>**
  - e.g. silicon wafer



(a) Gate



(b) Memory cell

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Two Metrics: Bandwidth and Latency

Bandwidth or throughput

- *Total work done in a given time*
- 10,000-25,000X improvement for processors
- 300-1200X improvement for memory and disks

Latency or response time

- *Time between start and completion of an event*
- 30-80X improvement for processors
- 6-8X improvement for memory and disks

Fundamentals

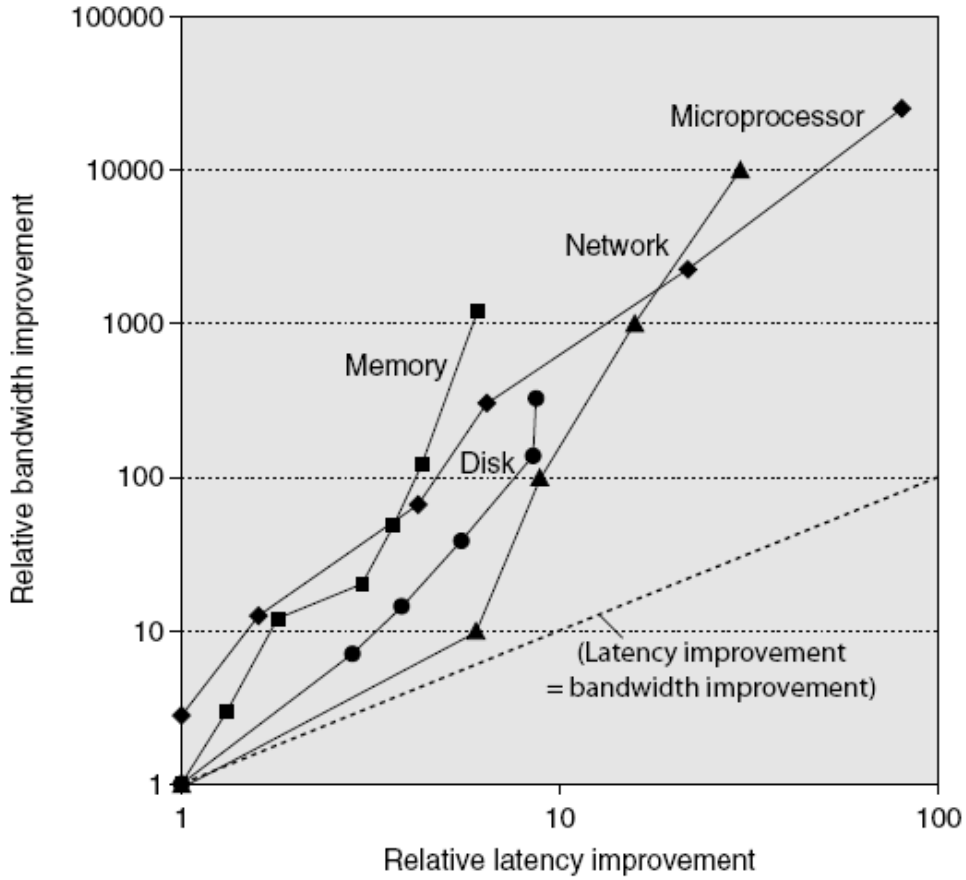
Microprocessor	16-bit address/ bus, microcoded	32-bit address/ bus, microcoded	5-stage pipeline, on-chip I & D caches, FPU	2-way superscalar, 64-bit bus	Out-of-order 3-way superscalar	Out-of-order superpipelined, on-chip L2 cache	Multicore OOO 4-way on chip L3 cache, Turbo
Product	Intel 80286	Intel 80386	Intel 80486	Intel Pentium	Intel Pentium Pro	Intel Pentium 4	Intel Core i7
Year	1982	1985	1989	1993	1997	2001	2010
Die size (mm^2)	47	43	81	90	308	217	240
Transistors	134,000	275,000	1,200,000	3,100,000	5,500,000	42,000,000	1,170,000,000
Processors/chip	1	1	1	1	1	1	4
Pins	68	132	168	273	387	423	1366
Latency (clocks)	6	5	5	5	10	22	14
Bus width (bits)	16	32	32	64	64	64	196
Clock rate (MHz)	12.5	16	25	66	200	1500	3333
Bandwidth (MIPS)	2	6	25	132	600	4500	50,000
Latency (ns)	320	313	200	76	50	15	4
Memory module	DRAM	Page mode DRAM	Fast page mode DRAM	Fast page mode DRAM	Synchronous DRAM	Double data rate SDRAM	DDR3 SDRAM
Module width (bits)	16	16	32	64	64	64	64
Year	1980	1983	1986	1993	1997	2000	2010
Mbits/DRAM chip	0.06	0.25	1	16	64	256	2048
Die size (mm^2)	35	45	70	130	170	204	50
Pins/DRAM chip	16	16	18	20	54	66	134
Bandwidth (MBytes/s)	13	40	160	267	640	1600	16,000
Latency (ns)	225	170	125	75	62	52	37
Local area network	Ethernet	Fast Ethernet	Gigabit Ethernet	10 Gigabit Ethernet	100 Gigabit Ethernet		
IEEE standard	802.3	803.3u	802.3ab	802.3ac	802.3ba		
Year	1978	1995	1999	2003	2010		
Bandwidth (Mbits/sec)	10	100	1000	10,000	100,000		
Latency (μsec)	3000	500	340	190	100		
Hard disk	3600 RPM	5400 RPM	7200 RPM	10,000 RPM	15,000 RPM	15,000 RPM	
Product	CDC WrenI 94145-36	Seagate ST41600	Seagate ST15150	Seagate ST39102	Seagate ST373453	Seagate ST3600057	
Year	1983	1990	1994	1998	2003	2010	
Capacity (GB)	0.03	1.4	4.3	9.1	73.4	600	
Disk form factor	5.25 inch	5.25 inch	3.5 inch	3.5 inch	3.5 inch	3.5 inch	
Media diameter	5.25 inch	5.25 inch	3.5 inch	3.0 inch	2.5 inch	2.5 inch	
Interface	ST-412	SCSI	SCSI	SCSI	SCSI	SAS	
Bandwidth (MBytes/s)	0.6	4	9	24	86	204	
Latency (ms)	48.3	17.1	12.7	8.8	5.7	3.6	

Milestones for

1) microprocessors,
2) memory,
3) networks, and
4) disks

Bandwidth and Latency

- Latency improved 6x to 80x
- Throughput improved 300x to 25,000x
- **Bandwidth has outpaced latency!**



Log-log plot of bandwidth and latency milestones

Feature Size and Its Impacts

Feature size is the minimum size of transistor or wire in x or y dimension

- **Integration density** scales quadratically with feature size
  - 10 microns in 1971 to .032 microns in 2011
- **Wire delay** does not scale well with feature size!
  - Wire delay is proportional to the product of <u>resistance</u> and <u>capacitance</u>

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Power and Energy

Problem: Get power in, get power or heat out

- ***Thermal Design Power (TDP)***
  - Characterizes sustained power consumption
  - Used as target for power supply and cooling system
  - Lower than peak power, higher than average power consumption

- **Clock rate** can be reduced dynamically to limit power consumption

Dynamic Energy and Power

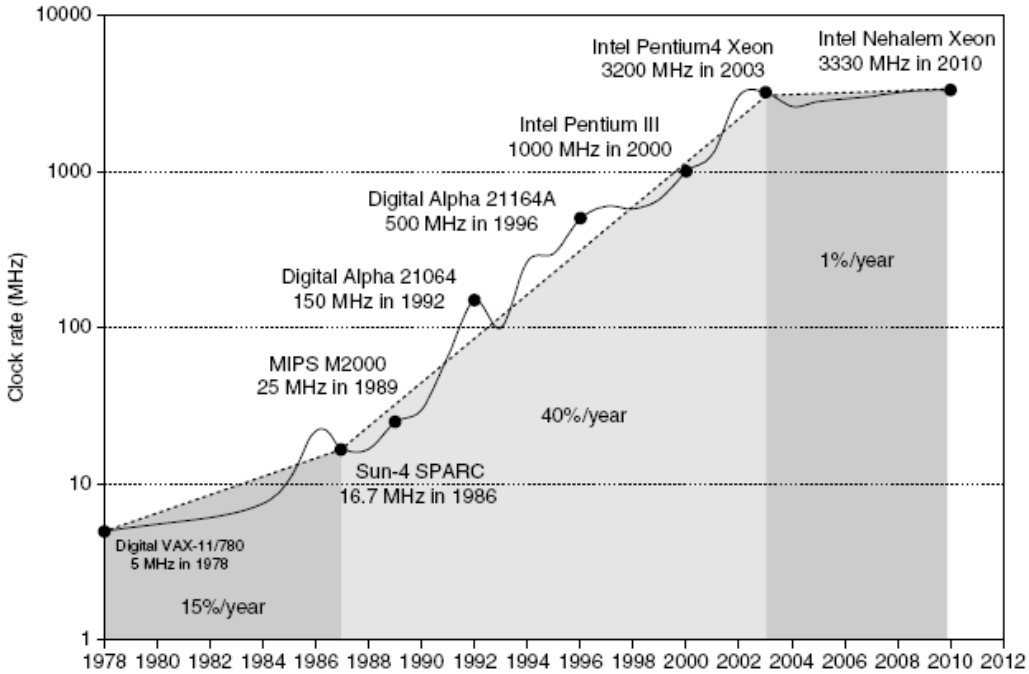
- ***Dynamic energy per transistor***
  - Used for a transistor switching from 0 -> 1 or 1 -> 0
  - ½ x Capacitive load x Voltage^2
  - The capacitive load is a function of *number of transistors* connected to output and *the technology* which determines the capacitance of wires and transistors
- ***Dynamic power*** *per transistor*
  - ½ x Capacitive load x Voltage^2 x Frequency switched

- **Voltage is the key**
  - The voltage of processors has become lower and lower

Power

- **Intel 80386** consumed ~ 2 W
- 3.3 GHz **Intel Core i7** consumes **130 W**
- Heat must be dissipated from 1.5 x 1.5 cm chip

- **This is the limit of what can be cooled by air**



Power vs. Energy

- **Power** is defined as the energy consumed in a unit time
- Question: Which metric should we use to compare processors: power or energy?

- It is better to use energy used for a specific task
  - Energy is tied to the a specific task and the time required for that task

Techniques for Reducing Power

Turn off the clock	• Turn off the clock of inactive modules
Dynamic Voltage-Frequency Scaling	• In periods of low activity, operate at lower frequency
Low power state for DRAM, disks	• In low power states, accesses are not allowed
Overclocking, turning off cores	• To turn off other cores and just run on a few cores

Static Power

- **Cause: The leakage current flows even when a transistor is off**

- $\text{Power}_{\text{static}} = \text{Current}_{\text{static}} \times \text{Voltage}$
  - Scales with number of transistors

- To reduce static power: ***power gating***, <u>turning off the power supply</u>